NOOR SYED

Lahore, Pakistan | noornsyed.14@gmail.com | +92 333 4898696 | LinkedIn | GitHub

Quantitative researcher with hands-on experience in large-scale field experiments, causal inference, and end-to-end ML pipelines. Built Transformer language models, Vision Transformers, diffusion models, and federated learning systems from scratch. Background spans Economics, Mathematics & CS — oriented toward research at the boundary of machine learning, consumer behavior, and digital markets.

EDUCATION

BSc (Hons.) Economics, Mathematics & Computer Science *Sep 2022 – May 2026*

Lahore University of Management Sciences (LUMS) | CGPA: 3.83 | SCGPA: 3.90 | Dean's Honor List (Top 5%)

Relevant Coursework: Deep Learning · Foundations of AI & ML · Principles & Techniques of Data Science · Econometrics I & II · Advanced Microeconomics · Game Theory · Linear Algebra · Probability & Statistics · Real Analysis · Operations Research I & II · Computational Problem Solving

RESEARCH EXPERIENCE

Research Data Scientist | LUMS Learning Institute *Sep 2025 – May 2026*

- Designed and executed a causal study using Propensity Score Matching (ML-based, Python & STATA) to evaluate the impact of attendance on student performance; managed full pipeline from survey design to quantitative analysis (5,000+ records).
- Built interactive dashboards and visualizations to communicate causal findings to academic and non-specialist stakeholders.
- Conducted systematic literature review under Dr. Hadia Majid, Dr. Ummad Mazhar, and Dr. Malik Jahan Khan.

Data Scientist – EdTech Hub (RCT, 560 Schools) | CERP / University of Chicago *Jun – Aug 2025*

Large-scale RCT evaluating a tech-supported foundational learning programme, under Prof. Christina Brown (University of Chicago).

- Engineered longitudinal data pipelines on 80,000+ observations, improving dataset merge rates from 47% to 95% — enabling scalable performance tracking across a multi-school field experiment.
- Applied Double Machine Learning (DRLearner / Random Forests) to estimate Average Treatment Effects, controlling for teacher fixed effects and confounders — methodology directly applicable to estimating causal effects of digital marketing and pricing interventions.
- Conducted heterogeneity analysis by performance quartile, gender, and class level (Bonferroni-corrected regressions); translated segment profiles into a data-driven resource allocation hierarchy — analogous to consumer segmentation for targeted marketing.
- Built interactive analytics dashboards (Power BI, Python) communicating causal findings to government stakeholders and non-technical decision-makers.

Data Analyst – WASH & Impact Evaluation | Tayaba Organization (with Reckitt) *Jun – Sep 2024*

- Designed and executed digital marketing and content strategy across social channels; used engagement analytics and audience data to optimize content decisions and reach — practitioner-level exposure to consumer behavior, digital advertising effectiveness, and content performance measurement.
- Managed impact evaluation and data pipeline for emergency-response programming reaching 3,986 beneficiaries; developed dashboards and a Situational Report for decision-makers.
- Co-authored a proposal recognised at COP28 and awarded the Impact Hero 2025 Award; synthesised analytical findings into policy-relevant narratives.

Research Assistant | University of Lahore, Dept. of Economics (HEC-funded) *Aug – Nov 2023*

- Analysed survey data from 11,650+ respondents (STATA, Python) across two HEC-funded projects; applied regression modelling to identify determinants of women's earnings; co-authored policy brief linked to SDG 5.

DEEP LEARNING & ML PROJECTS

NLP: Transformer Language Model from Scratch | Deep Learning Coursework *2026*

- Built a GPT-style causal language model end-to-end — custom tokenization (tiktoken/GPT-2), multi-head self-attention with causal masking, 4-layer Transformer with pre-LayerNorm, and autoregressive text generation. Relevant to consumer review analysis, voice search, and multimedia content modeling.

Computer Vision: Vision Transformer, CNNs & Generative Models | Deep Learning Coursework *2025–2026*

- Implemented Vision Transformer (ViT) from scratch with patch embedding, sinusoidal positional encodings, and CLS-token classification; extended to Hybrid CNN-ViT with convolutional patch stems on CIFAR-10. Applicable to image-based marketing data (Instagram, livestream video).
- Designed efficient CNN (289K params, 82.38% on CIFAR-10 under 1M budget); built GradCAM interpretability with PyTorch hooks to visualize model attention layer-by-layer; U-Net for dense image segmentation boosted Dice from 0.676 to 0.820.
- Built full diffusion model stack from scratch: DDPM (U-Net denoiser, cosine noise schedule), DDIM deterministic fast sampling (near-identical quality at 50 vs 1000 steps), and Latent Diffusion with VAE compression and classifier-free guidance.

Federated Learning: Heterogeneity, Fairness & Custom Aggregation | Deep Learning Coursework *2026*

- Built FedAvg and FedProx from scratch; deployed on State Farm distracted driver dataset (24 drivers, 10 classes) — FedProx nearly doubled worst-driver accuracy (37% → 67%).
- Designed FairServer, a novel fairness-aware aggregation algorithm using exponentiated gradient updates; achieved +9pp worst-client accuracy over FedAvg and 88% reduction in per-client accuracy variance.

ECONOMETRIC RESEARCH PROJECTS

Senior Thesis: Discipline & Human Capital Formation | LUMS | UNICEF MICS Data (N=15,590) *Aug–Dec 2025*

- Hybrid econometric–ML study on household disciplinary practices and early literacy/numeracy; ordered logit and generalized ordered logit with robustness checks; one-vs-rest Random Forest with SMOTE, SHAP, permutation importance, and LOCO validation. Semester CGPA: 3.9.

Time Series: Pakistan Demographic Dividend | Advanced Econometrics, LUMS *Aug–Dec 2025*

- VECM and mediation analysis in R on demographic structure's impact on GDP growth; bootstrapped impulse response functions identified weak capital formation as binding constraint.

TEACHING

Teaching Assistant – Principles & Techniques of Data Science | LUMS *Jan – Jun 2025*

- Designed and delivered Python-based data analysis training to 100+ undergraduates: pandas, scikit-learn, statsmodels, and applied ML pipelines.

TECHNICAL SKILLS

Languages & Data: Python (pandas, NumPy, statsmodels, scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM), R (tidyverse, ggplot2, VECM), SQL, STATA/SE, C++

Deep Learning & AI: Transformers (GPT, ViT, Hybrid CNN-ViT), Diffusion Models (DDPM, DDIM, Latent Diffusion, CFG), CNNs, U-Net, Autoencoders, RNNs, MLPs, Federated Learning (FedAvg, FedProx), GradCAM, SHAP, LOCO

NLP & Computer Vision: Language modeling, tokenization (tiktoken/GPT-2), attention variants (sparse, temperature-scaled), patch-based image modeling, image segmentation, GradCAM interpretability, PCA-CNN analysis

Causal Inference & Econometrics: RCT design & analysis, Double Machine Learning (DRLearner), Difference-in-Differences, Regression Discontinuity, Instrumental Variables, Propensity Score Matching, Panel Data, VECM, Heterogeneity Analysis

Experimentation & Segmentation: Large-scale field experiments, A/B testing, Qualtrics-compatible experimental design, K-Means & hierarchical clustering, treatment effect estimation, consumer segmentation

Data Engineering & Visualization: End-to-end pipeline development, web scraping, feature engineering, Power BI, Tableau, Matplotlib, Seaborn, ggplot2, QGIS, Excel/VBA

Game Theory: Strategic interaction models, Nash equilibrium, mechanism design — applicable to pricing and platform competition research

HONOURS & CERTIFICATIONS

25 Under 25 Award — Australian High Commission | Dean's Honor List, LUMS (Top 5%, multiple semesters) | Impact Hero 2025 Award

Google Data Analytics Professional Certificate | Responsible Conduct of Research (CITI) | Human Subjects Research – SBE (CITI)